

Lab 10 – Sets In Python

Programming Exercise:

1. Write a program which will add your best five students name in a set. You will use a loop to insert names in set.

Source Code:

```
1  # Initialize an empty set
2  best_students = set()
3
4  # Loop to take input for five students' names
5  for i in range(5):
6      student_name = input(f"Enter the name of student {i + 1}: ")
7      best_students.add(student_name) # Add the student name to the set
8
9  # Print the set of best students
10 print("The best five students are:", best_students)
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe"
Enter the name of student 1: ALI
Enter the name of student 2: ALIYAA
Enter the name of student 3: AHMED
Enter the name of student 4: AYAN
Enter the name of student 5: RAFAY
The best five students are: {'AHMED', 'ALIYAA', 'ALI', 'AYAN', 'RAFAY'}

Process finished with exit code 0
|
```

2. Write a program which will remove 2 friends who left NED

Source Code:

```
1 friends = {"ALI", "BISMA", "ALIYAA", "DANISH", "ESHAAL"}
2 print("Initial friends at NED:", friends)
3 # Remove two friends who left NED
4 friends.discard("DANISH")
5 friends.discard("ALI")
6 print("Updated friends at NED:", friends)
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\PycharmProje
Initial friends at NED: {'ESHAAL', 'DANISH', 'ALI', 'ALIYAA', 'BISMA'}
Updated friends at NED: {'ESHAAL', 'ALI', 'ALIYAA', 'BISMA'}
```

```
Process finished with exit code 0
```

3. Write a program which will add your best dishes and then pop one by one until the set is empty.

Source Code:

```
1 best_dishes = {"Pizza", "Pasta", "Sushi", "Burger", "Tacos"}
2 print("Best dishes:", best_dishes)
3 # Pop dishes one by one until the set is empty
4 while best_dishes:
5     dish = best_dishes.pop()
6     print(f"Popped dish: {dish}")
7 # Print the final empty set
8 print("Final set of dishes (should be empty):", best_dishes)
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Use
Best dishes: {'Sushi', 'Pizza', 'Burger', 'Tacos', 'Pasta'}
Popped dish: Sushi
Popped dish: Pizza
Popped dish: Burger
Popped dish: Tacos
Popped dish: Pasta
Final set of dishes (should be empty): set()

Process finished with exit code 0
```

4. Write a program which will store number of items in a set after each purchasing the items will be pop from the set and compare its price at the end program will give you the total amount of items have been sold. Also find the max amount and minimum amount of items sold.

Source Code:

```
1  # Initialize an empty set for storing item prices
2  item_prices = set()
3
4  # Ask the user for the number of items to be purchased
5  num_items = int(input("How many items are you purchasing? "))
6
7  # Input prices of the items and add them to the set
8  for i in range(num_items):
9      price = float(input(f"Enter the price of item {i + 1}: "))
10     item_prices.add(price)
11
12  # Variables to track the total, max, and min prices
13  total_amount = 0
14  max_price = float('-inf') # Start with a very low value for max price
15  min_price = float('inf')  # Start with a very high value for min price
16
```

```
17     # Pop items from the set and calculate total, max, and min prices
18     while item_prices:
19         price = item_prices.pop() # Remove an arbitrary item (price)
20         total_amount += price # Add the price to the total amount
21         max_price = max(max_price, price) # Update max price if necessary
22         min_price = min(min_price, price) # Update min price if necessary
23
24     # Print the results
25     print(f"Total amount of items sold: {total_amount}")
26     print(f"Maximum price of an item sold: {max_price}")
27     print(f"Minimum price of an item sold: {min_price}")
28
```

Output:

C:\Users\hp\PycharmProjects\main\venv\Scripts\p

How many items are you purchasing? 5

Enter the price of item 1: 20.5

Enter the price of item 2: 45.6

Enter the price of item 3: 36

Enter the price of item 4: 78.9

Enter the price of item 5: 23

Total amount of items sold: 204.0

Maximum price of an item sold: 78.9

Minimum price of an item sold: 20.5

Process finished with exit code 0

5. Write a program which will compare two sets, Set A and Set B. Both the sets have some students who love to play one is hockey and other one is cricket. 10 of them play both. Now using sets find how many of them are playing cricket only, if universal set is 40, students who play hockey are 21

Source Code:

```
1  # Given data
2  total_students = 40  # Universal set
3  students_hockey = 21  # Students who play hockey
4  students_both = 10  # Students who play both cricket and hockey
5
6  # Calculate the size of Set B (students who play cricket)
7  students_cricket = total_students - (students_hockey - students_both)
8
9  # Students who play cricket only (Set B - Set A)
10 cricket_only = students_cricket - students_both
11
12 # Print the result
13 print(f"The number of students who play cricket only is: {cricket_only}")
```

Output:

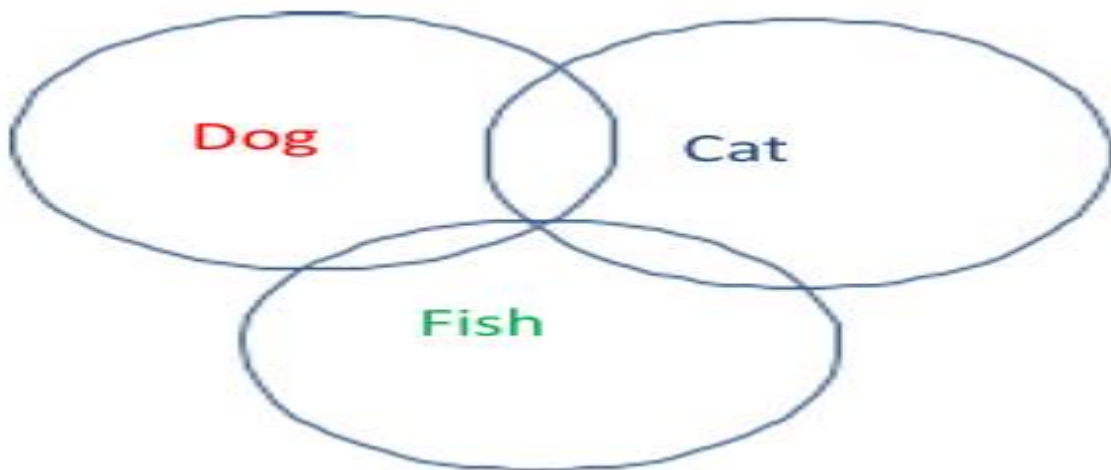
```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\l
The number of students who play cricket only is: 19
```

```
Process finished with exit code 0
```

6. A pet store keeps track of the purchases of customers over a four-hour period. The store manager classifies purchases as containing a dog product, a cat product, a fish product, or product for a different kind of pet. She found.

- a. 83 purchased a dog product
- b. 101 purchased a cat product
- c. 22 purchased a fish product
- d. 31 purchased a dog and a cat product
- e. 8 purchased a dog and a fish product
- f. 10 purchased a cat and a fish product
- g. 6 purchased a dog, a cat and a fish product
- h. 34 purchased a product for a pet other than a dog, cat or a fish.

- i. How many purchases were for a dog product only?
- ii. How many purchases were for cat product only?
- iii. How many purchases for a dog or a fish product?
- iv. How many purchases were there in total?



Source Code:

```
1  # Given Data
2  dog = 83
3  cat = 101
4  fish = 22
5  dog_cat = 31
6  dog_fish = 8
7  cat_fish = 10
8  dog_cat_fish = 6
9  other_pets = 34
10
11  # Total customers (universal set)
12  total_customers = 40
13
14  # i. How many purchases for a dog product only
15  dog_only = dog - dog_cat - dog_fish + dog_cat_fish
16
17  # ii. How many purchases were for cat product only
18  cat_only = cat - dog_cat - cat_fish + dog_cat_fish
19
20  # iii. How many purchases for a dog or a fish product
21  dog_or_fish = (dog + fish - dog_fish - dog_cat_fish)
22
23  # iv. How many purchases were there in total (including other pets)
24  total_purchases = dog + cat + fish - dog_cat - dog_fish - cat_fish + dog_cat_fish + other_pets
25
26  # Display the results
27  print(f'i. Dog product only: {dog_only}')
28  print(f'ii. Cat product only: {cat_only}')
29  print(f'iii. Dog or fish product: {dog_or_fish}')
30  print(f'iv. Total purchases: {total_purchases}')
```

Output:

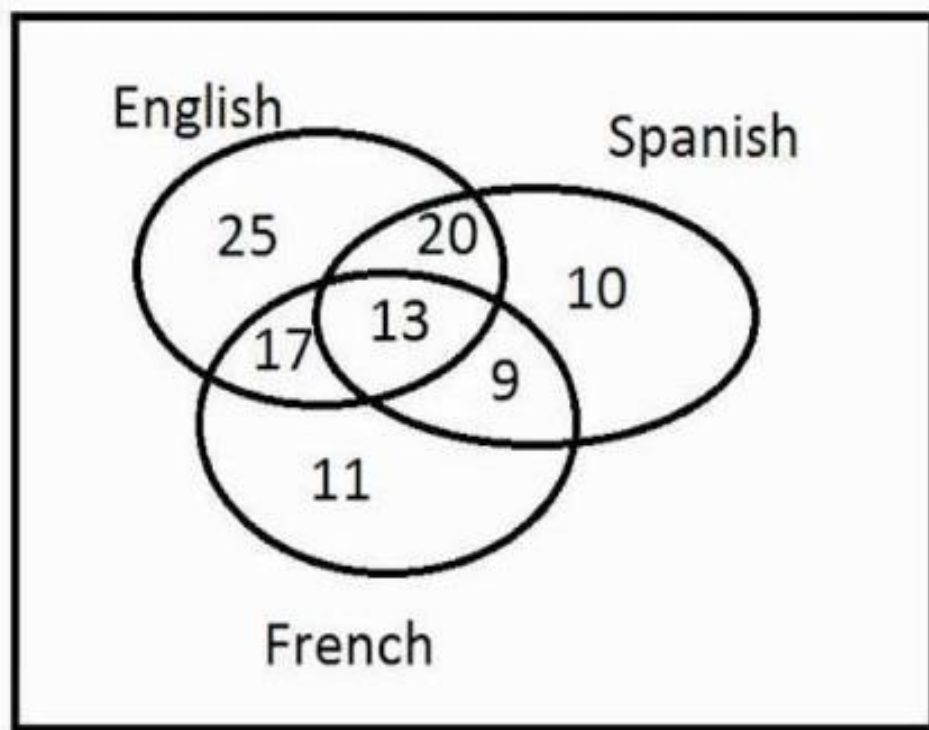
C:\Users\hp\PycharmProjects\main\venv

```
i. Dog product only: 50
ii. Cat product only: 66
iii. Dog or fish product: 91
iv. Total purchases: 197
```

Process finished with exit code 0

7. Solve the following problem of real world. A camp of international students has 110 students, as shown in the diagram. The diagram will elaborate that all the students speak some kind of a language. We need to find out how many that speak none of them out of 110 students. Find how many students speak

- a. English and Spanish but not French?
- b. Neither English, Spanish, nor French?
- c. French, but neither English nor Spanish?
- d. Only one of the three languages?
- e. Exactly two of the three languages?



As a programmer your task is to verify that above Venn diagram is correctly filled.

Source Code:

```
1  # Data from the diagram
2  english_only = 25
3  spanish_only = 20
4  french_only = 11
5  english_spanish_not_french = 13
6  english_french_not_spanish = 7
7  spanish_french_not_english = 9
8  all_three = 13
9
10 # Total number of students
11 total_students = 110
12
13 # Students who speak at least one language
14 students_one_or_more = (english_only + spanish_only + french_only + english_spanish_not_french
15                        + english_french_not_spanish + spanish_french_not_english + all_three)
16
17 # a. English and Spanish but not French
18 a = english_spanish_not_french
19
20 # b. Neither English, Spanish, nor French
21 b = total_students - students_one_or_more
22
23 # c. French, but neither English nor Spanish
24 c = french_only
25
26 # d. Any one of the three languages
27 d = english_only + spanish_only + french_only
28
29 # e. Exactly two of the three languages
30 e = english_spanish_not_french + english_french_not_spanish + spanish_french_not_english
31
32 # Display the results
33 print(f"a. English and Spanish but not French: {a}")
34 print(f"b. Neither English, Spanish, nor French: {b}")
35 print(f"c. French, but neither English nor Spanish: {c}")
36 print(f"d. Any one of the three languages: {d}")
37 print(f"e. Exactly two of the three languages: {e}")
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\
```

- a. English and Spanish but not French: 13
- b. Neither English, Spanish, nor French: 12
- c. French, but neither English nor Spanish: 11
- d. Any one of the three languages: 56
- e. Exactly two of the three languages: 29

```
Process finished with exit code 0
```